KiniBoku

An Infrastructure Layer for the Informal Economy

Hackathon Architecture & Pitch Document
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1. Project Overview & Problem Statement

The Problem: The local informal economy—comprising artisans, mechanics, thrift sellers, and street food vendors—operates almost entirely on chaotic, unstructured data. These traders rely on fleeting WhatsApp statuses, Instagram stories, and informal word-of-mouth. For consumers, this makes finding and verifying reliable local services incredibly difficult. Standard search engines cannot parse a voice note or a picture of a market stall, and it is notoriously hard to separate genuine traders from scammers hiding behind temporary accounts.

The Solution: KiniBoku is an AI-driven infrastructure layer that transforms this unstructured social media noise into a clean, searchable, and verified map of local commerce. By utilizing multi-modal LLMs to onboard vendors via simple WhatsApp voice notes and images, and employing vector databases for semantic search, KiniBoku connects the invisible informal economy directly to consumers and formal financial systems.

2. Hackathon Theme Alignment: Challenge 02

Squad Hackathon 3.0: Smart Systems - The Intelligent Economy

KiniBoku is a direct response to **Challenge 02**: Designing an intelligent economic system powered by data and AI that connects informal traders and financial services in one ecosystem. We view financial inclusion not as a charity problem, but as a technology problem.

- **AI Automation:** Bypassing complex forms by automating vendor onboarding through NLP and audio transcription via WhatsApp.
- **Use of Data:** Generating predictive "Trust Scores" using alternative data signals (social media velocity, sentiment analysis) rather than traditional credit histories.
- **Squad API Integration:** Squad APIs serve as the core transactional layer. Payments between users and informal traders are processed via Squad, turning informal interactions into documented, secure financial histories.

- **Financial Innovation:** Using social traction and cross-platform consistency to build verified digital and financial identities for the unbanked.

3. Core Feature Implementation

Frictionless, Multi-Modal Onboarding

Informal traders lack the time or digital literacy for extensive profile forms. KiniBoku bypasses this friction by utilising a WhatsApp bot as the primary vendor interface.

- **Mechanism:** Traders send unstructured data (e.g., a voice note saying "I fix generators around Akoka" or a photo of their stall).
- **Backend Processing:** Small Language Models (SLMs) transcribe the audio and analyse images to automatically extract the vendor's category, location, and services.

Generalised Social Scraping

The scraping engine is designed to categorise a massive variety of unstructured social media posts, capturing the breadth of the informal market.

- **Transactional Keyword Expansion:** Searches for transactional intent (e.g., "DM for price," "Delivery nationwide") rather than specific product names.

Conversational "Fuzzy" Search

Users interact with the database via natural language, bridging the gap between colloquial search terms and vendor descriptions.

- **Semantic Matching:** Uses a vector database to match the intent of a query (e.g., "cracked screen repair") with extracted vendor profiles.
- **Location Anchoring:** Results are geographically anchored, ranking informal services by proximity using scraped location tags.

4. The Popularity System: Trust & Traction

To provide social proof without falling victim to easily manipulated vanity metrics, KiniBoku implements a composite Trust & Traction Score. This metric algorithmically separates genuine market activity from bot-driven engagement.

$$T_s = (L \times w_1) + (C_{sent} \times w_2) + (V_{freq} \times w_3)$$

- L (Likes): High volume, but low weight (w_1).
- C_{sent} (Comment Sentiment): SLM-driven sentiment analysis filters for meaningful engagement over generic emojis (w_2).
- V_{freq} (Velocity/Frequency): Consistent posting activity over a prolonged period indicates a legitimate business operation (w_3).

5. Security & Vendor Verification

Addressing fraud risks in peer-to-peer informal markets requires non-traditional verification.

The "Cross-Platform Consistency" Index

KiniBoku performs a historical cross-check across scraped data. If a vendor's phone number or brand name appears consistently across multiple platforms over a minimum span of six months, they are awarded a "Verified Legacy" status.

The "Community Vouch" Chain

Leveraging communal trust networks, verified vendors can vouch for new entrants. If a vouched vendor is subsequently flagged for fraud, the vouching vendor's Trust Score is penalised. This creates a self-policing trust ecosystem.

6. The Hackathon Pitch: The "Wow Factor"

> **The Narrative:** KiniBoku is not merely a directory; it is an infrastructure layer. It tackles the core of the Intelligent Economy by applying sophisticated engineering bizarreness to the chaotic, unstructured noise of social media, transforming it into a clean, searchable, and verified map of local commerce powered by Squad.